

Project 6: Market Data Analysis and Strategy Screening Tool

Objective: Build a tool that organizes historical market data for multiple securities and screens them based on trading signals from different strategies.

Requirements:

1. Create a market data system using dictionaries and lists:
 - Store OHLC (Open, High, Low, Close) data for at least 5 different stocks
 - Include at least 20 days of price history for each stock
 - Include additional metrics like volume and calculated indicators (e.g., 5-day moving average)
2. Implement three simple trading strategies:
 - Moving average crossover (short MA crosses above/below long MA)
 - Momentum (positive/negative returns over N days)
 - Price breakout (price crosses above/below recent high/low)
3. For each strategy, generate buy/sell signals for each stock based on the historical data
4. Use appropriate data structures:
 - Lists for time series data like price histories
 - Tuples for immutable data points like specific OHLC values for a given day